Assignment #4

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Problem D.4, D.6

Find:
- Equilibria. (eqba.)
- Jacobian linearization. (J.L.)
- Feedback linearization (F.L.)

- For $\vec{x} = \begin{bmatrix} z \\ \dot{z} \end{bmatrix}$ and $\vec{u} = [F]$ and $\vec{y} = [z]$, find state space form (SSF):

$$\dot{\vec{x}} = A\vec{x} + B\vec{u}$$
$$\vec{y} = C\vec{x} + D\vec{u}$$

Solution

From Assignment #3, we have

$$\dot{\vec{x}} = \vec{f}(x, u) = \begin{bmatrix} d\dot{z}/dt \\ \frac{d}{dt}(\dot{z}) \end{bmatrix} = \begin{bmatrix} \dot{z} \\ \frac{1}{m}(F - b\dot{z} - kz) \end{bmatrix}$$

Then for $f(x_e, u_e) = 0 \ldots$

$$\dot{z}_e = 0$$

$$F_e - b\cancel{\dot{z}_e}^{\,0} - k z_e = 0 \Rightarrow F_e = k z_e$$

Therefore the equilibria are coupled such that...

$$\left.\begin{aligned} \dot{z}_e &= 0 \\ F_e &= k z_e \end{aligned}\right\} \leftarrow \text{equilibrium}$$

Note the EoMs are already linear, so linearization is unnecessary. ← JL FL

By inspection...

$$\left.\begin{aligned} \dot{\vec{x}} &= \begin{bmatrix} 0 & 1 \\ -k/m & -b/m \end{bmatrix} \begin{bmatrix} z \\ \dot{z} \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} [F] \\ \vec{y} &= [1 \quad 0] \begin{bmatrix} z \\ \dot{z} \end{bmatrix} + [0][F] \end{aligned}\right\} \leftarrow \text{SSF}$$

Problem E.4, E.6

Find: Sameas D, with $\vec{\tilde{x}} = \begin{bmatrix} \tilde{z} \\ \tilde{\theta} \\ \dot{\tilde{z}} \\ \dot{\tilde{\theta}} \end{bmatrix}$ and $\vec{\tilde{u}} = [\tilde{F}]$

and $\vec{\tilde{y}} = \begin{bmatrix} \tilde{z} \\ \tilde{\theta} \end{bmatrix}$

From Assignment #3...

$$\dot{\vec{x}} = \vec{f}(\vec{x}, \vec{u}) = \begin{bmatrix} \frac{d}{dt}(z) \\ \frac{d}{dt}(\theta) \\ \frac{d}{dt}(\dot{z}) \\ \frac{d}{dt}(\dot{\theta}) \end{bmatrix} = \begin{bmatrix} \dot{z} \\ \dot{\theta} \\ -g\sin\theta + z\dot{\theta}^2 \\ \dfrac{3\left((2F\ell - m_2 g\ell - 2m_1 g z)\cos\theta - 4m_1 z\dot{\theta}\dot{z}\right)}{2(m_2\ell^2 + 3m_1 z^2)} \end{bmatrix}$$

Then equilibria when $\vec{f}(\vec{x}_e, \vec{u}_e) = \vec{0}$...implies...

$\dot{z}_e = 0$

$\dot{\theta}_e = 0$

$-g\sin\theta_e = 0 \Rightarrow \theta_e = \pi n$ for $n \in \mathbb{Z}$

$$\frac{3\left((2F_e\ell - m_2 g\ell - 2m_1 g z)\cos\theta_e\right)}{2(m_2\ell^2 + 3m_1 z_e^2)} = 0 \Rightarrow 2F_e\ell - m_2 g\ell - 2m_1 g z_e = 0$$

$\Rightarrow F_e\ell = \frac{1}{2}m_2 g\ell + m_1 g z_e$

$\Rightarrow F_e = \frac{1}{2}m_2 g + m_1 g \frac{z_e}{\ell}$

Therefore equilibria occur at...

$$\left.\begin{array}{ll} \dot{z}_e = 0 & \theta_e = \pi n \text{ for } n \in \mathbb{Z} \\ \dot{\theta}_e = 0 & F_e = \frac{1}{2}m_2 g + m_1 g \, z_e/\ell \end{array}\right\}$$ eqba.

Then by solving the jacobian of $\dot{\vec{x}}$ at equilibrium $\vec{x}_e = [\frac{\ell}{2}, 0, 0, 0]^T$ with $\vec{u}_e = [F_e] = [\frac{1}{2}(m_1 + m_2)g]$...

$$\dot{\vec{\tilde{x}}} = \begin{bmatrix} \dot{\tilde{z}} \\ \dot{\tilde{\theta}} \\ -g\tilde{\theta} \\ \dfrac{12\tilde{F}}{\ell(3m_1 + 4m_2)} - \dfrac{12 g m_1 \tilde{z}}{\ell^2(3m_1 + 4m_2)} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -g & 0 & 0 \\ \dfrac{-12 g m_1}{\ell^2(3m_1 + 4m_2)} & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \tilde{z} \\ \tilde{\theta} \\ \dot{\tilde{z}} \\ \dot{\tilde{\theta}} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \dfrac{12}{\ell(3m_1 + 4m_2)} \end{bmatrix} [\tilde{F}]$$

JL

Feedback linearization requires a forcing term to be in every non-linear EoM. However, $-g\sin\theta + z\dot{\theta}^2$ has no forcing term, so feedback linearization is not possible. FL

For state space form, see the Jacobian linearization for the first equation. The second is given (by inspection) as... SSF

$$\vec{\tilde{y}} = \begin{bmatrix} \tilde{z} \\ \tilde{\theta} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \tilde{z} \\ \tilde{\theta} \\ \dot{\tilde{z}} \\ \dot{\tilde{\theta}} \end{bmatrix} + [0][\tilde{F}]$$

Problem F.4, F.6

Find: Same as D, with $\vec{u} = \begin{bmatrix} F \\ \tau \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ d & -d \end{bmatrix} \begin{bmatrix} f_r \\ f_l \end{bmatrix} = \begin{bmatrix} f_r + f_l \\ d(f_r - f_l) \end{bmatrix}$.

This implies $f_r = \frac{1}{2}F + \frac{1}{2d}\tau$ and $f_l = \frac{1}{2}F - \frac{1}{2d}\tau$.

$\vec{x} = [z_v, h, \theta, \dot{z}_v, \dot{h}, \dot{\theta}]^T$ and $\vec{y} = [z_v, h]^T$.

From assignment #3...

$$\dot{\vec{x}} = \vec{f}(\vec{x}, \vec{u}) = \begin{bmatrix} d/dt(z_v) \\ d/dt(h) \\ d/dt(\theta) \\ d/dt(\dot{z}_v) \\ d/dt(\dot{h}) \\ d/dt(\dot{\theta}) \end{bmatrix} = \begin{bmatrix} \dot{z}_v \\ \dot{h} \\ \dot{\theta} \\ \dfrac{-((f_l+f_r)\sin\theta + \mu\dot{z}_v)}{m_c + 2m_r} \\ \dfrac{(f_l+f_r)\cos\theta - g(m_c+m_l+m_r)}{m_c+2m_r} \\ \dfrac{d(f_r-f_l)}{J_c + 2d^2 m_r} \end{bmatrix} = \begin{bmatrix} \dot{z}_v \\ \dot{h} \\ \dot{\theta} \\ \dfrac{-F\sin\theta - \mu\dot{z}_v}{m_c + 2m_r} \\ \dfrac{F\cos\theta - g(m_c+m_l+m_r)}{m_c+2m_r} \\ \dfrac{\tau}{J_c + 2d^2 m_r} \end{bmatrix}$$

Therefore equilibria occur when $\vec{f}(\vec{x}_e, \vec{u}_e) = 0$...

$\dot{z}_{ve} = 0$ $\quad -F_e\sin\theta_e = 0 \Rightarrow F_e = 0$ or $\theta_e = n\pi$ for $n \in \mathbb{Z}$

$\dot{h}_e = 0$ $\quad F_e\cos\theta_e = g(m_c + m_l + m_r) \Rightarrow F_e = \dfrac{g(m_c+m_l+m_r)}{\cos\theta_e} \neq 0$

$\dot{\theta}_e = 0$ $\quad \tau_e = 0$

Therefore equilibria occur for...

$\dot{z}_{ve} = 0$ $\quad \theta_e = 0$ $\quad \theta_e = n\pi$ for $n \in \mathbb{Z}$ $\quad z_{ve}, h_e \in \mathbb{R}$

$\dot{h}_e = 0$ $\quad \tau_e = 0$ $\quad F_e = \cos\theta_e\, g(m_c + m_l + m_r)$ $\quad \forall$ — eqba

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Then selecting point $\vec{x}_e = [0, 0, 0, 0, 0, 0]^T$ and $\vec{u}_e = [(m_c + m_l + m_r)g, \; 0]^T$ as the equilibrium to evaluate the Jacobian, sympy yields...

$$\dot{\vec{\tilde{x}}} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & \frac{-g(m_c+m_l+m_r)}{m_c+2m_r} & \frac{-\mu}{m_c+2m_r} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \tilde{z}_v \\ \tilde{h} \\ \tilde{\theta} \\ \dot{\tilde{z}}_v \\ \dot{\tilde{h}} \\ \dot{\tilde{\theta}} \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \frac{1}{m_c+2m_r} & 0 \\ 0 & \frac{1}{J_c+2d^2m_r} \end{bmatrix} \begin{bmatrix} \tilde{F} \\ \tilde{\tau} \end{bmatrix}$$

JL

Feedback linearization is no possible because there are 2 equations that are non-linear and only one associated input variable, F. Additionally, F is attached to the nonlinear term, which makes it impossible also without setting $F = 0$.

FL

For SSF, see JL above for $\dot{\vec{\tilde{x}}}$. By inspection,

$$\vec{y} = \begin{bmatrix} \tilde{z} \\ \tilde{h} \\ \tilde{\theta} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \tilde{z}_v \\ \tilde{h} \\ \tilde{\theta} \\ \dot{\tilde{z}}_v \\ \dot{\tilde{h}} \\ \dot{\tilde{\theta}} \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \tilde{F} \\ \tilde{\tau} \end{bmatrix}$$

Note that rows 2 and 5 give longitudinal solution and rows 1, 3, 4 & 6 give the lateral states. Corresponding columns of A would likewise be neglected.

SSF